Lipoplex surface targeting — wet-lab handout

Which receptor to put on the particle, which antibody to order, and what to check first.
Prepared 12 August 2026 · computational assessment, no wet-lab data · supersedes nothing in `antibody_assessment.md`, extends it.

> **Bottom line**
>
> Use **TfR + P2X3** as the targeting pair. Drop **TfR + TRPV1** — TRPV1 has almost no extracellular surface for an antibody to bind on an intact cell. **No P2X3 antibody currently on the market is validated for live-cell surface binding.** The first experiment is therefore a surface-binding test, not a formulation study.

1 · Why P2X3 and not TRPV1

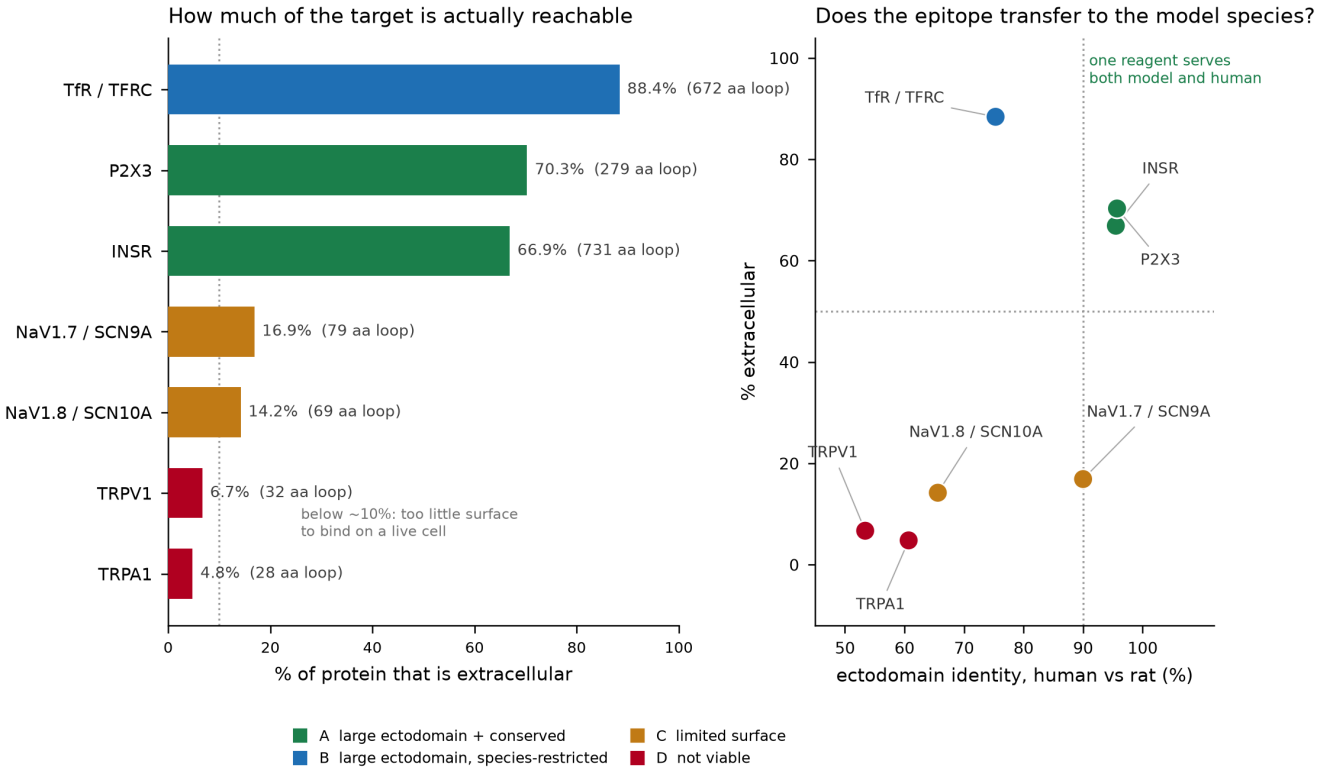
An antibody on a particle can only grip the part of a receptor that sticks out of the cell. We measured that fraction for every target named in the project bibliography, using UniProt membrane topology, and asked separately whether the exposed part is conserved between rat and human (which decides whether one reagent serves both the animal model and eventual human work).

Target	Role	% outside the cell	Largest loop (aa)	Identity, whole protein	Identity, exposed part	Tier
TfR / TFRC	BBB crossing (established)	88.4%	672	76.7%	**75.3%**	B
INSR	BBB crossing (alternative)	66.9%	731	94.4%	**95.5%**	A
P2X3	Nociceptor targeting	70.3%	279	94.0%	**95.7%**	A
NaV1.7 / SCN9A	Nociceptor targeting	16.9%	79	93.1%	**90.0%**	C
NaV1.8 / SCN10A	Nociceptor targeting	14.2%	69	83.2%	**65.6%**	C
TRPV1	Nociceptor targeting	6.7%	32	85.8%	**53.4%**	D
TRPA1	Nociceptor targeting	4.8%	28	79.7%	**60.7%**	D

Identity is human vs rat. Tiers: **A** large exposed surface and conserved · **B** large surface, species-restricted · **C** limited surface · **D** not a viable antibody surface.

> **The two numbers that decide it**
>
> P2X3 presents **70.3%** of itself outside the cell, in a single continuous 279-residue loop. TRPV1 presents **6.7%**, split across four loops of which the largest is 32 residues.
>
> And where every failing target is less conserved on its exposed face than overall (TRPV1 drops 32 points), P2X3 goes the other way — its exposed face is **95.7%** identical between human and rat, better than the protein average. Note that TfR itself is only 75.3%, which is exactly why anti-TfR clones cannot be swapped between species.

Lipoplex surface targets: P2X3 is the only nociceptor marker with a bindable surface



Left: how much of each target is reachable from outside. Right: the same targets against how well the exposed part is conserved between species.

2 · Antibody shortlist for P2X3

Update — better reagents found after this handout's first draft

A registry-wide screen (103 P2X3 records) found two reagents whose published immunogen sequence was verified against the ectodomain, which the leads below were not: **ABclonal A10232 (AB_2757757)**, immunogen aa 44–317 — essentially the whole 44–322 ectodomain, listed for rat, human 95.6% identical over those positions; and **Atlas HPA057776 (AB_2683526)**, immunogen aa 184–260, listed for human, rat 92.2% identical. Together they cover both species with verified extracellular epitopes. Order these alongside PA5-77682. Neither is validated for live-cell binding — the binding gate in §3 still applies. Separately, the **Bioss bs-4249R** series is the only P2X3 reagent listing flow cytometry and ships pre-conjugated (14 variants), but its epitope is undisclosed — request it.

Screened two ways: mining 99 open-access P2X3 antibody papers for cited RRIDs, and a direct registry query returning all **103 P2X3 records (93 unique reagents)**. Eight are listed below. Each vendor epitope was checked against the P2X3 sequence — the ectodomain is **aa 44–322** of 397.

Clone / catalogue	Target	Species reactivity	Verdict	Notes and what to check
PA5-77682 (polyclonal) AB_2736292	P2X3 (rat/ mouse) extracellular	mouse, rat	**LEAD CANDIDATE - rodent only; validate before use**	The only registry reagent explicitly raised against the P2X3 EXTRACELLULAR domain. Vendor applications are ICC/IF, WB, IHC(F) - live-cell surface labelling is NOT a listed application and appears in 0 of 97 open-access surface/live-cell papers screened. Reacts mouse/rat only: not usable for human work. Polyclonal, so no orientation control for conjugation. Immunogen sequence not disclosed in registry - request it from the vendor and check it falls inside aa 44-322 before ordering.

Clone / catalogue	Target	Species reactivity	Verdict	Notes and what to check
APR-016 AB_2313760	P2X3 (h/m/r)	human, mouse, rat	**DO NOT USE for targeting - epitope is CYTOPLASMIC**	Most-cited P2X3 antibody in the field. Vendor epitope (C)VEKQSTDSGAYSIGH maps to aa 383-397, which is C-terminal cytoplasmic (ectodomain is 44-322) - it cannot bind an intact cell. Also note the listed peptide is the RAT sequence; human P2X3 383-397 is AEKQSTDSGAFSIGH (13/15 identical). Fine for WB/IHC on permeabilized tissue; unusable as a targeting ligand.
PA5-115707 AB_2900342	P2X3	human, mouse, rat	**BACKUP - epitope undisclosed, confirm before use**	Broadest species reactivity of the P2X3 reagents found (h/m/r), which matters given the ectodomain is 95.7% conserved human-vs-rat. Registry lists no epitope and no surface/flow application (ICC/IF, IHC-P, WB). Request the immunogen sequence and confirm it lies within aa 44-322.
RA10109 antiserum AB_2157931	P2X3	not stated	**CAUTION - flow listed but epitope undisclosed**	Registry comments list flow cytometry among applications, but give no epitope and no indication whether flow was on permeabilized cells. Whole antiserum, not purified IgG - unsuitable for conjugation without affinity purification.
H68.4 AB_2533029	TfR (h/m/r + others)	human, mouse, rat, hamster, chicken	**NOTE - binds cytoplasmic tail, not for targeting**	Found co-cited in the P2X3 literature scan. Broad species reactivity, but H68.4 is raised against the TfR cytoplasmic domain - it is a detection reagent, not a transcytosis ligand. Does not replace OX-26 / RI7217 for targeting.

Read this before ordering anything

The most-cited P2X3 antibody in the literature, **Alomone APR-016**, is raised against aa 383–397 — that is the **cytoplasmic tail**. It cannot bind an intact cell and is useless as a targeting ligand, however many papers use it. It is a perfectly good western-blot and permeabilized-IHC reagent, which is what those papers use it for.

Separately: the epitope Alomone lists is the **rat** sequence. Human P2X3 differs at two of those fifteen positions (`AEKQSTDSGAFSIGH` vs `VEKQSTDSGAYSIGH`).

3 · What to do first

1. **Request immunogen sequences** from Thermo for **PA5-77682** and **PA5-115707**. Ask for the peptide, not the datasheet. Check it falls inside aa 44–322. Do not order until you have it.
2. **Live-cell binding test.** P2X3-expressing line (or rat DRG primary culture) against a P2X3-negative control line. Stain **unpermeabilized**, on ice, with a viability dye to gate out dead cells — dead cells take up antibody non-specifically and are the usual source of a false positive here. Read by flow cytometry. This is the gate: if nothing binds the intact cell, the targeting strategy stops here.
3. **Permeabilized control in parallel.** Same antibody, same cells, permeabilized. If it stains permeabilized but not intact cells, the epitope is internal — that is an informative negative, not a failed experiment.
4. **Only then** conjugate to the particle and repeat the binding test on the formulated lipoplex.

4 · Honest limitations

- **P2X3 has never been used as a nanoparticle targeting ligand.** Literature counts: 283 papers on P2X3 as a nociceptor marker, 52 on clinical antagonists (the gefapixant programme), but only 9 touching live-cell surface antibody work and 13 any nanoparticle targeting. The biology is solid; the targeting application is new. Budget for validation, not for citation.
- **Topology is annotation, not structure.** The single large P2X3 loop is consistent with the known trimeric P2X architecture, but whether a given epitope is occluded by glycosylation or by the neighbouring subunit is not answered here.

- **95.7% identity over 279 residues still permits ~12 substitutions.** A specific clone's epitope could sit on one of them. This analysis reliably rules targets out; it does not guarantee any particular clone transfers between species.
- **P2X3 is not exclusively neuronal.** It is expressed on bladder urothelium and in the airway — the mechanism behind gefapixant's cough indication and its taste side-effects. A P2X3-targeted particle that reaches systemic circulation has off-target tissue.
- **Marker ≠ compartment.** As already noted in the project bibliography, a nociceptor marker is not the same thing as a central pain region. Whether P2X3+ peripheral terminals are the right compartment for this delivery route is a separate question this analysis does not touch.

5 · Sources

Topology and sequence: UniProt reviewed entries, resolved by gene symbol and organism (P2X3 `P56373` / `P49654` / `Q3UR32`; TRPV1 `Q8NER1`; TRPA1 `O75762`; SCN9A `Q15858`; SCN10A `Q9Y5Y9`; INSR `P06213`; TFRC `P02786`). Identity by global Needleman–Wunsch alignment. Reagents: Antibody Registry records resolved by RRID, cross-referenced against Europe PMC open-access full texts. Literature counts: PubMed, 12 August 2026. Full data in `lipoplex_targeting_assessment.csv` and `antibody_shortlist_p2x3.csv`.